PneumoScan AI

Chest X-ray screening report · Pneumonia detection using ResNet

Scan ID PS-20261107-0DFC08
Date 07 Nov 2026, 23:39
Source file pneumonia_01.jpeg

Patient

PATIENT NAME	AGE	SEX	PATIENT ID
Priya Raman	58	Female	OP-2026-0142

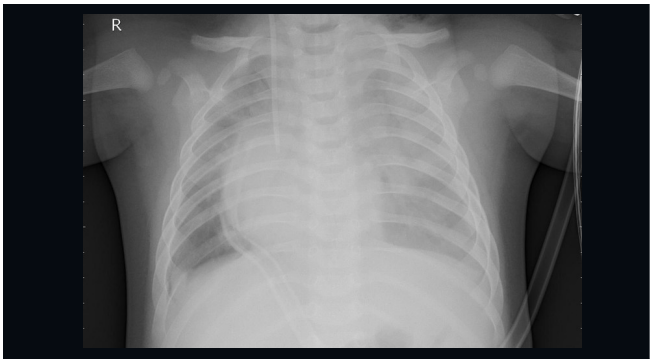
SYMPTOMS
Cough and fever for 4 days, short of breath on walking

AI result

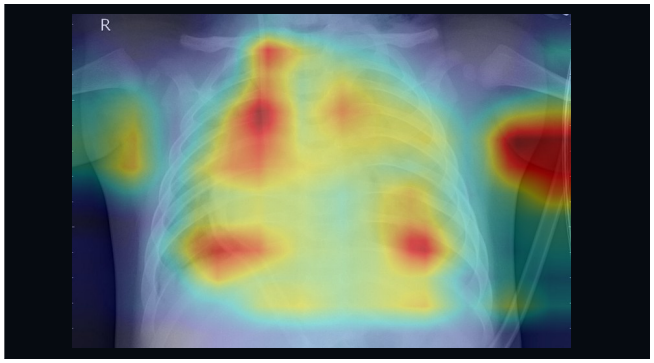
● Pneumonia likely

The model found patterns in this X-ray that it learned to associate with pneumonia. The heatmap shows the regions that most influenced this decision. This is a screening signal for a doctor to review, not a diagnosis. The decision threshold is set low on purpose: the model is tuned to rarely miss pneumonia, so borderline cases are flagged for a doctor to review.

RISK LEVEL	PNEUMONIA PROBABILITY	DECISION THRESHOLD	TEST-SET SENSITIVITY
High	**95.6%**	**7.4%**	**99.7%**



Chest X-ray as analysed by the model



Grad-CAM heatmap: warmer colours mark the regions that most influenced the result

About the model

Model version pneumoscan-resnet-ensemble-v1
Held-out test set (624 X-rays): accuracy 94.6% · sensitivity 99.7% · specificity 85.9% · ROC-AUC 0.987
Inference time 639 ms
The model was trained and tested on paediatric chest X-rays (ages 1–5) from a single hospital in Guangzhou, China. It may perform worse on adults, other hospitals or other X-ray machines.

Doctor's notes

Right lower zone opacity on review.
Started oral antibiotics; review in 48 h.

Reviewing doctor _____ Signature and date _____

Recommended next steps

- Show this result and the original X-ray to a qualified doctor. Only a doctor can confirm or rule out pneumonia.
- Doctors confirm pneumonia with the history, a physical exam and tests such as pulse oximetry and blood tests.
- For adults, the doctor may use a severity score such as CURB-65 to decide where care should happen.
- Call an ambulance (India: 108, or 112) for struggling to breathe, pale, blue or blotchy skin, lips or tongue, coughing up blood, or sudden confusion.

Disclaimer. This report was produced by PneumoScan AI, a student research project that analyses chest X-ray images with a deep-learning model. It is not a certified medical device and does not provide a diagnosis. All results must be reviewed and confirmed by a qualified medical professional before any clinical decision is made. If you feel very unwell or have trouble breathing, seek emergency care immediately.